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Investigating parameters affecting the real and virtual drapability of silk fabrics for traditional Hanbok

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Abstract

This paper presents research into the parameters affecting the real and virtual drapability of silk fabrics for traditional Hanbok. The KES-FB (Kawabata evaluation system of fabric) and the CLO (3D fashion design software) fabric kit system were used to evaluate the physical properties. The relationships between physical properties and the drape coefficient of silk fabrics were statistically analyzed. It was determined whether the objectified real fabric data could be used and applied as a database when implementing virtual clothing using a 3D virtual clothing program. The KES-FB properties and CLO fabric kit results indicated that the bending property and weight were significantly correlated with the drape coefficient. The bending rigidity and weight per unit area were influential parameters of the drape coefficient. The regression analysis results of both the KES-FB and the CLO fabric kit revealed that bending rigidity and thickness were statistically significantly correlated with the drape coefficient. The bending property was the most determining parameter for the drape coefficient of silk fabrics. A statistically significant difference between the real and virtual drape coefficients of stiffer silk fabrics was found. In the CLO 7.1 program, the accuracy of drape implementation of virtual fabrics was found to be useful when the bending stiffness of the actual fabric was less than 0.089 gf-cm²/cm and the elasticity was more than 1.03%. The results revealed that more research on the program modeling method considering the mechanical and physical properties of the real fabric and its structure is needed.

Keywords: Real and virtual drapability, Silk fabrics, Traditional Hanbok, KES-FB system, 3D CLO system

Introduction

The commercialization of digital technology through three-dimensional (3D) technology convergence has presented the opportunity to transform the fashion industry into a next-generation realistic platform. 3D virtual clothing digital technology is becoming an important factor in creating competitiveness in clothing design, manufacturing, production, and marketing (Sayem, 2015). Recently, as non-face-to-face services have been strengthened in the fashion industry, 3D virtual clothing systems have been actively introduced to e-commerce, digital fashion shows, showrooms, and virtual fittings due to

the strength of online markets (Plotkina & Saurel, 2019). The implementation of virtual clothing using a 3D computer-assisted design (CAD) program is done by users who are skilled in the program and have accumulated experience so that the realistic expression of fabric properties can be expressed. Therefore, experts familiar with computers and programs are required for sophisticated fabric simulation (Buyukaslan, 2017). Improvement in such realistic expression depends on accurate physical property expression of the fabrics. Specialized physical property measurement equipment must be used to apply the physical properties of a specific fabric to virtual clothing. The disadvantages of a 3D virtual wearing system include the limitation of fabric expression, which does not represent the texture or drape of real fabric well, with differences between real and virtual fabrics (Choi & Nam, 2009). Realistically expressing the silhouette of 3D virtual clothing so that the virtual fabric can accurately reflect the characteristics of the real fabric, with drape and silhouette of the virtual fabric similar to those of the real fabric, is very important (Choi, 2022; Portfield & Lamar, 2017; Rudolf et al., 2016). Drapability is the 3D deformation of fabrics caused by gravity. It is an important factor that defines the appearance of clothing, such as color, gloss, and texture (Pattanayak et al., 2010). The drapability of fabrics differs depending on fiber composition, thread, fabric structure, and finishing type, with each fabric having its own drape shape (Abdin et al., 2013; Glombikova & Zdenek, 2014; Lozen & Jevsnik, 2007; Sanade et al., 2013).

In apparel products, the aesthetic perception inherent in the textile is expressed according to the mechanical properties of the textile material. Fabric drapability is a major factor that constitutes the appearance and silhouette of clothing, along with pattern composition (Kenkare & May-Plumlee, 2005). Hu and Chen (1998) investigated the relationship between physical properties and the overall drape coefficient using the Kawabata estimating system (KES-F) and found that the drape coefficient was closely related to the bending and shear properties, the mean deviation of the friction coefficient, and tensile linearity. Niwa and Seto (1986) found that the drape coefficient of women's dress fabrics was greatly affected by $\{2HB \text{ (hysteresis width at a bending curvature)}/W \text{ (weight per unit area)}\}^{1/2}$ among properties such as bending stiffness, bending hysteresis, weight per unit area, shear stiffness, and shear hysteresis. Morooka and Niwa (1976) also studied drape in which the drape coefficient was correlated with mechanical properties, according to the statistical analysis, a high correlation was obtained between multi regression models of drape ratio-bending property-weight. Matsudaira and Yang (2000) proposed that bending rigidity, bending hysteresis, weight per unit area, shear stiffness, and shear hysteresis were closely related to the drape coefficient of new synthetic fabrics. Although the KES-F can measure the mechanical properties of fabrics very accurately and reproducibly, it also has disadvantages in that it is expensive, making it challenging for general users or individual designers to access this equipment (Chan et al., 2006; Ju & Choi, 2020; Tsukada et al., 2013). As such, the characteristics of fabrics are important factors constituting clothing.

The need for research on the physical properties of real and virtual fabrics for developing 3D virtual clothing systems is increasing. 3D virtual clothing programs include "3D Runway Designer", developed by Optitex in Israel, "V-Stitcher" and "Lotta", developed by Browzwear in Singapore, "I-Designer", developed by Technoa in Japan, and "3D-Fit", developed by Lectra in France. "DC-Suite", developed by Picsen Co. Ltd.,

“Qualoth”, developed by FXGear, and “CLO 3D”, developed by Clo Virtual Fashion Co., Ltd., are 3D virtual clothing programs developed in Korea (Kim et al., 2014a, 2014b; Lage & Ancutiene, 2017; Power, 2013; Sayem et al., 2010). The CLO 3D virtual clothing program used in this study is a 3D virtual clothing CAD program developed by CLO Virtual Fashion Co., Ltd. It is suitable for the fashion industry and was started from a program called “Marvelous Designer”, which could realistically express game and animation character costumes (Na & Kim, 2012).

Clothing-related studies that are being conducted using 3D virtual clothing programs include studies that compare and evaluate the appearance of virtual clothes and real clothes at the same time and studies that evaluate fit and suggest optimal patterns using virtual clothing systems (Fontana et al., 2005). Studies belonging to this category include a comparative study of men’s classic fit and slim fit shirt patterns (Kim et al., 2014a, 2014b), a comparison of real clothing with online and 3D virtual garments (Kim & Na, 2013), and pattern corrections of women’s tailored jackets (Kwak, 2016). Most of these preceding studies were comparative studies of changes in patterns or differences in the appearance images of clothing. Since the expressive power of virtual clothing is very closely related to the physical properties of fabrics in a 3D virtual clothing system, studies (Buyukaslan et al., 2017; Lee & Kim, 2016) have proposed improvements in programs related to the expression of fabric properties. Differences between the results of real and virtual clothing were seen when virtual clothing was simulated by applying physical properties measured in real fabrics and compared to a 3D virtual clothing program (Yoon, 2022). However, such studies are insufficient.

Hanbok is traditional clothing that embodies the Korean lifestyle and culture. The form and structure of Hanbok have changed in various ways, depending on the style of life, culture, living environment, and aesthetics of the times (joint operation of related ministries, 2021). Unlike Western clothing, which is designed to fit a 3D shape, Hanbok is designed using a flat pattern and changes to suit the body shape of the wearer. High-quality silk fabric has been used throughout Korean history to make Hanbok for the royal family and people of high social status. Currently, silk fabric is used as a clothing material for Hanbok for ceremonies, such as weddings, in the general public. Recently, global interest in Korean contents (K-contents) and Korean culture has heightened interest in traditional Hanbok clothing. As Korean pop (K-pop) culture has grown in popularity around the world, Hanbok has become an inspiration to all K-pop fans who first discovered it on YouTube (Victoria & Albert Museum. London, 2023). Research on the real and virtual drapability of traditional Hanbok silk fabrics for the utilization of virtual garments is needed to promote the popularity of Korean-culture.

Therefore, in this study, 30 silk fabrics used as textiles for Hanbok, a traditional Korean costume, were selected to compare the physical properties of real and virtual fabrics. KES-FB (Kato Tech Co., Ltd) properties, the CLO fabric kit properties, and the drapability of silk fabrics were measured, and correlations were analyzed to investigate the effects of the physical properties of real fabric and virtual fabric on drapability. Quantified and objectified property data on the silk fabrics are expected to be used to build a digital textile database to produce 3D virtual clothing.

Methods

Materials

The experimental fabrics used in this study were provided by the Korea Silk Research Institute. Thirty types of silk fabrics were 100% silk fabric used in Korean women's clothes (Hanbok), with a weight of 3.705 ~ 11.588 mg/cm² (average weight: 6.927 mg/cm²) and a thickness of 0.176 ~ 0.557 mm (average thickness: 0.284 mm) range. Properties of these 30 types of silk fabrics are summarized in Table 1. Scanned images of 30 silk fabrics are presented in Fig. 1.

Mechanical properties of silk fabrics by KES-FB

The commonly used the KES-FB system was used in this study to obtain mechanical properties of silk fabrics. A square sample of 20 cm × 20 cm in size was used for measurement. As shown in Table 2, physical properties of the fabric were measured three times in warp and weft directions under standard conditions for 17 items of six properties: tensile, bending, shearing, compression, surface properties, weight and thickness (Kawabata, 2000).

Physical properties of silk fabrics by CLO fabric kit

To obtain physical properties using the CLO fabric kit, a 22 cm × 3 cm rectangular sample was prepared in warp, weft, and bias directions, respectively. As measurement items, characteristics of tensile stiffness, bending stiffness, shear stiffness, weight, and thickness were measured as shown in Table 3. Measurements were repeated three times. When measuring physical properties, the test was conducted in the order of weight, thickness, bending, and tensile characteristics with less damage. Samples in the warp, weft, and bias directions required to measure weight were folded and put on the scale to measure the weight. Weight per unit area was obtained by dividing weight by the fabric area. The thickness was measured with a digital thickness gauge. The contact distance and the bending length were obtained by testing one warp and one weft swatch as bending stiffness. The stretch stiffness was measured in five stages with the force corresponding to the length of the fabric sample fixed at both ends. Shear stiffness corresponded to the tensile stiffness measurement of a fabric sample cut in the bias direction. Measured values were entered in the emulator within the CLO and digitalized.

Drape coefficient

A visual image drape meter was used to measure the drape coefficient (Kim, 2011; Park et al., 2004). This method adds a digital image processor to the widely used Cusick drape tester (Cusick, 1968). First, the drape shape was obtained using a Cusick drape tester according to KS K ISO 9073-9 (Korean Agency for Technology & standards, 2021). A sample with a diameter of 30 cm was placed on a cylinder with a disk having a diameter of 18 cm. The shape of the draped sample was photographed by setting the distance between the drape measuring device and the camera lens to 76 cm. The drape coefficient was obtained by calculating the area of the draped shape by obtaining a two-dimensional shape using the image analysis system from data obtained through the image input

Table 1 Characteristics of silk fabrics

Fabric No.	Fabric contents	Weave	Fabric density (warp × weft/ in ²)	Weight (mg/cm ²)	Thickness (mm)
s1	Raw Silk 100%	Plain	112 × 76	3.705	0.311
s2	Raw Silk 100%	Jacquard	166 × 76	4.705	0.203
s3	Raw Silk 100%	Plain	148 × 72	4.718	0.288
s4	Raw Silk 100%	Plain	160 × 68	4.735	0.241
s5	Raw Silk 100%	Jacquard	100 × 68	4.785	0.215
s6	Degummed Silk 100%	Jacquard	98 × 77	4.968	0.229
s7	Raw Silk 100%	Jacquard	128 × 82	5.095	0.176
s8	Raw Silk 100%	Plain	90 × 64	5.435	0.272
s9	Raw Silk 100%	Jacquard	204 × 102	5.515	0.220
s10	Raw Silk 100%	Plain	188 × 72	5.705	0.228
s11	Degummed Silk 100%	Plain	214 × 68	5.870	0.217
s12	Raw Silk 100%	Unbalanced plain	104 × 36	5.960	0.557
s13	Raw Silk 100%	Jacquard	188 × 84	6.015	0.252
s14	Degummed Silk 100%	Jacquard	324 × 92	6.328	0.213
s15	Raw Silk 100%	Plain	216 × 160	6.518	0.272
s16	Raw Silk 100%	Unbalanced plain	110 × 71	6.560	0.438
s17	Degummed Silk 100%	Plain	96 × 63	6.843	0.221
s18	Degummed Silk 100%	Jacquard	144 × 64	6.880	0.319
s19	Degummed Silk 100%	Jacquard	240 × 108	7.120	0.229
s20	Degummed Silk 100%	Plain	70 × 64	7.515	0.246
s21	Raw Silk 100%	Jacquard	112 × 112	7.880	0.327
s22	Degummed Silk 100%	Jacquard	184 × 74	8.148	0.249
s23	Degummed Silk 100%	Jacquard	300 × 114	8.395	0.234
s24	Degummed Silk 100%	Jacquard	308 × 132	8.440	0.241
s25	Degummed Silk 100%	Jacquard	302 × 112	8.590	0.296
s26	Degummed Silk 100%	Jacquard	225 × 104	9.755	0.332
s27	Degummed Silk 100%	Jacquard	292 × 128	9.855	0.316
s28	Degummed Silk 100%	Jacquard	224 × 108	9.998	0.397
s29	Degummed Silk 100%	Unbalanced plain	236 × 106	10.200	0.359
s30	Degummed Silk 100%	Jacquard	278 × 130	11.588	0.411

device. The drapability of the fabric was measured a total of six times on the front and back sides of three samples. The formula for obtaining the drape coefficient is shown in Eq. (1):

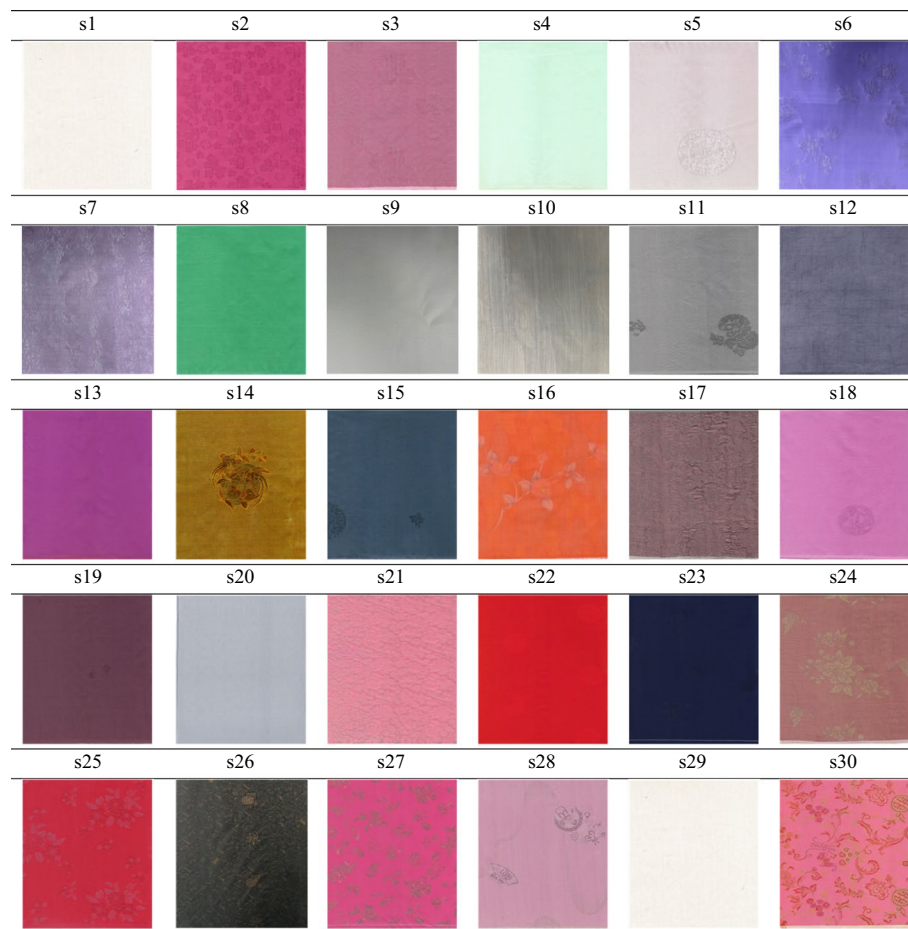


Fig. 1 Scanned images of 30 silk fabrics

Table 2 Characteristic values of KES-FB mechanical properties

Parameter		Description	Unit
Tensile	LT	Linearity of tensile property	None
	WT	Tensile energy per unit area	gf·cm/cm ²
	RT	Tensile resilience	%
	EM	Extensibility, strain at 500gf/cm of tensile load	%
Bending	B	Bending rigidity per unit length	gf·cm ² /cm
	2HB	Hysteresis of bending moment per unit length	gf·cm/cm
Shear	G	Shear stiffness	gf/cm·deg
	2HG	Hysteresis of shear force at 0.5° shear angle	gf/cm
	2HG5	Hysteresis of shear force at 5° shear angle	gf/cm
Compression	LC	Linearity of compression	None
	WC	Compressional energy	gf·cm/cm ²
	RC	Compressional resilience	%
Surface	MIU	Coefficient of friction	None
	MMD	Mean deviation of MIU	None
	SMD	Geometrical roughness	μm
Thickness	T	Thickness at 0.5 gf/cm ²	mm
Weight	W	Weight per unit area	mg/cm ²

Table 3 Characteristic values of CLO fabric kit physical properties

Parameter	Description	Unit
Tensile	Stretch stiffness-weft	g/s ²
	Stretch stiffness-warp	g/s ²
Shear	Stretch stiffness-bias	g/s ²
Bending	Bending stiffness-weft	g mm ² /s ² rad
	Bending stiffness-warp	g mm ³ /s ² rad
	Bending stiffness-bias	g mm ² /s ² rad
Thickness	Thickness at 1 ± 0.01kpa	mm
Weight	Weight per unit area	g/m ²

$$\text{Drape coefficient (\%)} = \{(A_2 - A_0)/(A_1 - A_0)\} \times 100(\%) \quad (1)$$

where,

A_0 = area of the supporting plate

A_1 = area of the specimen

A_2 = projected area by the specimen

Implementation of virtual drape image using the CLO 3D CAD

3D virtual drape shapes of 30 types of silk fabric were simulated using a 3D clothing production software CLO 3D CAD (CLO Virtual Fashion Co., Ltd., V.7.1). Physical properties measured with the CLO fabric kit were input into the emulator and digitized. They were converted into a virtual fabric with properties such as stretch stiffness-weft, stretch stiffness-warp, shear, bending stiffness-weft, bending stiffness-warp, bending stiffness-bias, thickness, and density. Shear corresponded to a measure of the stretch stiffness of a fabric sample cut in the bias direction. Thus, the actual value was reflected as it is. Buckling is an item reflecting characteristic that the fabric bends when more than a certain force is applied while maintaining its shape when an external force is applied. In this study, the buckling value were applied according to the type of silk fabric fixed in program. of the type of silk fabric fixed in the program was applied. Using this virtual fabric, the drape shape was simulated in the same way as the real drape measurement method. After opening the “Table OBJ” file in the “File” menu, the 18 cm virtual object circular plate with the same diameter as the real drape for measuring instrument’s supporting plate was run. Then a 30 cm circle pattern was created in the pattern window tool. A virtual fabric image was then created and applied as shown in Fig. 2. The formula for obtaining the virtual drape coefficient is shown in Eq. (2):

$$\text{Virtual drape coefficient (\%)} = \{(VA_2 - VA_0)/(VA_1 - VA_0)\} \times 100(\%) \quad (2)$$

where,

VA_0 = virtual area of the supporting plate

VA_1 = virtual area of the specimen

VA_2 = virtual projected area by the specimen

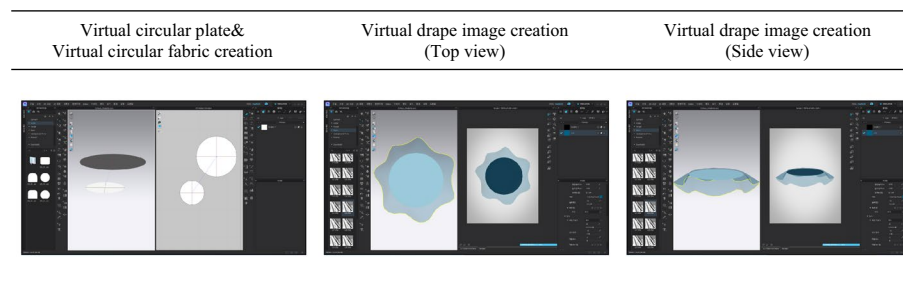


Fig. 2 Virtual drape image creation process in 3D CLO system

Statistical analysis

Statistical analysis was conducted with the IBM SPSS Statistics 26 program. The KES-FB and CLO fabric kit evaluation systems were analyzed for correlation. Correlation analysis was conducted between the drape coefficient, KES-FB properties, and CLO fabric kit measurements. In addition, for drape coefficient, multiple regression analysis was performed with KES-FB properties and CLO fabric kit measurements as independent variables. K-means cluster analysis was performed to analyze the drapability by grouping silk fabrics with similar drape coefficients. Difference between the two characteristics was compared and evaluated using an independent t-test and correlation analysis between the actual drape coefficient and the virtual drape coefficient.

Results and Discussion

Comparison of measurement results by KES-FB and CLO fabric kit

The first step of this study was to compare the two evaluation systems. As shown in Fig. 3, the correlation coefficient between the KES-FB properties and the CLO fabric kit measurement results was as high as 0.964** (p-value < 0.01) in the case of weight. In addition, in the case of thickness, the correlation coefficient was 0.787** and the bending property was 0.518**. These values showed similar correlations although the two measurement systems used different measurement principles. However, no significant correlation was derived for tensile property. In the case of thickness and weight, there was a slight difference between the values measured using the two systems (CLO fabric kit and KES-FB). The thickness (value) measured using the KES-FB system tended to be slightly larger than that measured using the CLO fabric kit. This means that the thickness measurement of the KES-FB system refers to the thickness measured at the moment when the pressure plate comes down and touches the sample and receives force. It is the thickness (value) measured when the force (0.5 gf/cm²) applied to the sample is close to 0. On the other hand, when measuring with the CLO fabric kit's digimatic thickness gauge, the thickness is measured under a greater pressure (1 ± 0.01 kPa), so it appears to be slightly thinner than the thickness (value) measured with the KES-FB system. There appears to be a slight difference in weight values due to the difference in unit area applied to weight measurement in the two systems. In the case of tensile property, when measuring using the CLO fabric kit, the length of increase and the force applied at that time are measured point by point, so there is no continuity. However, when measuring using the KES-FB system, the force applied as the sample is stretched is continuously measured.

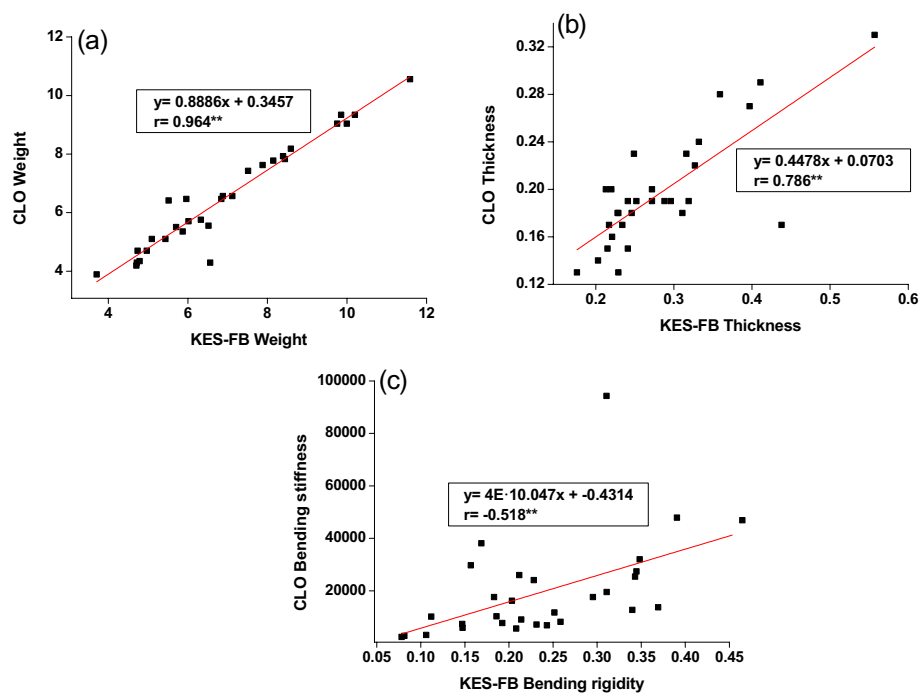


Fig. 3 Comparison of KES-FB and CLO fabric kit parameters: **a** Weight, **b** Thickness, and **c** Bending property

Additionally, in the KES-FB system, the maximum force was set at 500 gf/cm, but in the CLO fabric kit system, the experiment was conducted without setting the maximum force. Therefore, the difference in measured values between the two systems appears to be due to differences in data collection and experimental methods. According to Yoon's research (Yoon, 2021a), when comparing the correlation between the physical properties measured with the CLO fabric kit and the KES-FB system, the weight, thickness, and bending stiffness showed a high correlation of over 0.9. But the tensile property showed a very low correlation of 0.33. As mentioned in previous studies (Pattanayak et al., 2010; Power, 2013; Shyr et al., 2007), the stretch stiffness value of the CLO fabric kit showed the same result as the low correlation coefficient between physical property values of virtual and real fabric in the case of fabrics having a very low elasticity.

Correlation and regression analysis between KES-FB properties and drapability of silk fabrics

Figure 4 shows results of the correlation analysis between drape coefficient and KES-FB mean value properties. The drape coefficient showed significant negative correlations with characteristics of extensibility (-0.503^{**}) (p -value < 0.01), and weight (-0.402^{*}) (p -value < 0.05). In addition, drape coefficient showed significant positive correlations with characteristics of tensile resilience (0.432^{*}), bending rigidity (0.431^{*}) and mean deviation of MIU(Coefficient of friction) (0.533^{**}). The extensibility representing elongation deformation when a maximum load of 500 gf/cm was applied showed a result consistent with a previous study. It was found that the more difficult the stretch, the higher the drape coefficient. In addition, the greater the linearity caused by weight and compression, the smaller the drape coefficient (Sung et al., 1987). Komatsu and Niwa (1981)

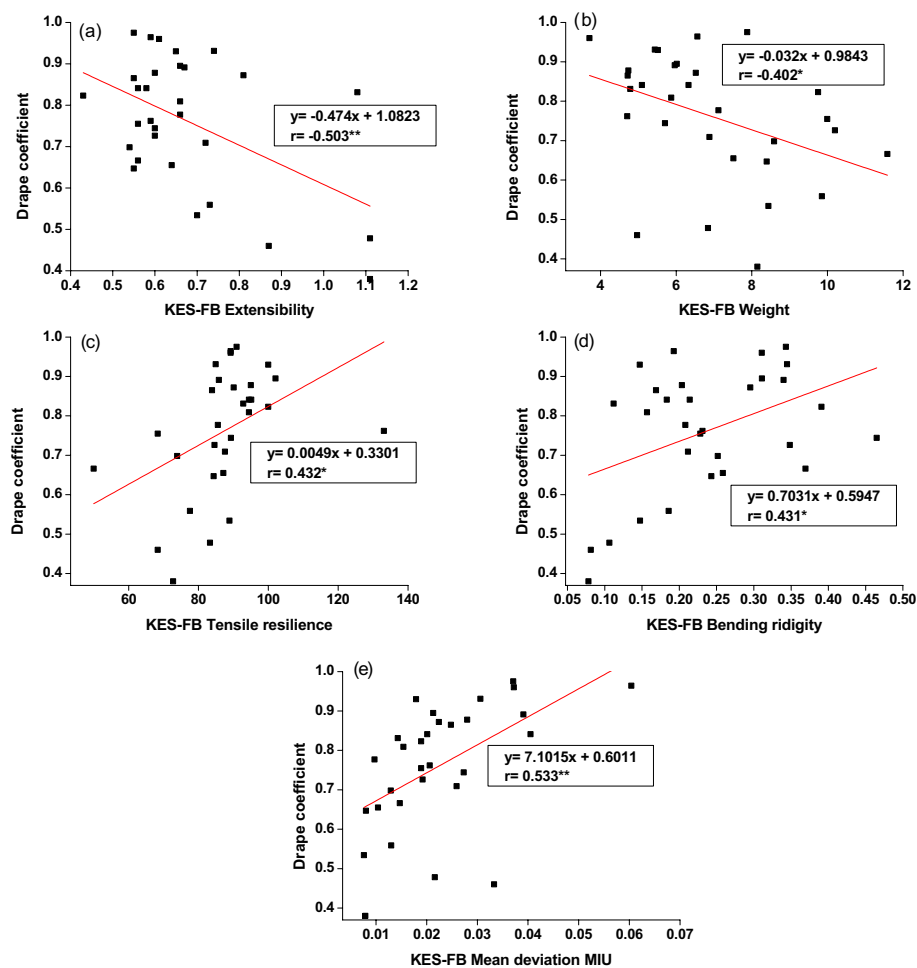


Fig. 4 Comparisons of drape coefficient and KES-FB parameters: **a** Extensibility, **b** Weight, **c** Tensile resilience, **d** Bending rigidity, **e** Mean deviation of MIU

have reported that the greater the tensile resilience, the greater the shape retention. They also found that when the bending rigidity had a large value, the space was maintained from the body and a box-shaped silhouette was formed. Hu and Chan have suggested that mean deviation of the friction coefficient is closely related to the drape ratio of a fabric (Hu & Chan, 1998). That is, when Mean deviation of MIU (MMD) representing resistance to slip is large, it means that the irregularity of the surface is large. This result is related to the drape ratio of the fabric.

Multiple regression analysis was performed between drape coefficient and KES-FB properties. As shown in Table 4, r^2 of the regression model was about 78.6%. Among variables as independent variables, bending rigidity, thickness, compressional energy, tensile energy and hysteresis width at shear angle 0.5° were influential variables that could explain drape coefficient of the fabrics.

Table 4 Regression analysis results of KES-FB variables for drape coefficient of silk fabrics

Independent variables	Unstandardized coefficients		Standardized coefficients	t	r ²
	B	Standard error	β		
(Constant)	0.541	0.247		2.186*	
KES-FB Tensile energy	− 0.547	0.139	− 0.381	− 3.936**	
KES-FB Bending rigidity	1.397	0.273	0.856	5.120***	0.786
KES-FB Hysteresis width at shear angle 0.5°	0.185	0.074	0.449	2.491*	
KES-FB Compressional energy	17.099	5.023	1.163	3.404**	
KES-FB Thickness	− 1.652	0.604	− 0.865	− 2.736**	

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

The regression equation of this regression model was expressed in Eq. (3) as follows:

$$\begin{aligned}
 \text{Drape coefficient} = & 0.541 - 0.547 \times (\text{KES} - \text{FB Tensile energy}) \\
 & + 1.397 \times (\text{KES} - \text{FB Bending rigidity}) + 0.185 \\
 & \times (\text{KES} - \text{FB Hysteresis width at shear angle } 0.5^\circ) \\
 & + 17.099 \times (\text{KES} - \text{FB Compressional energy}) \\
 & - 1.652 \times (\text{KES} - \text{FB Thickness})
 \end{aligned} \quad (3)$$

Tokmak et al. (2010) also studied drape in which the drape coefficient was correlated with mechanical properties, a high correlation was obtained between multi regression models of drape ratio-bending-shear property and the additional tightness factor. Collier (1991), and Shyr et al. (2007) have also performed a step-by-step multiple regression analysis to study the relationship between drape coefficient and the physical properties, the bending and shearing properties were found to be most closely associated with the static drape coefficient for the test fabrics. To determine multicollinearity (Hair et al., 1998), which represents the relationship between three or more variables, tolerance and VIF (variance inflation factor) checks were performed. The tolerance check revealed that KES-FB bending rigidity, thickness, compressional energy, tensile energy and hysteresis width at shear angle 0.5° had values above 0.1. The VIF revealed that these variables had values below 10. Therefore, it could be said that multicollinearity between independent variables in this regression model was not a problem. In addition, the above independent variables used for the regression equation had a p-value of less than 0.05, meaning that they had significant effects on the dependent variable. In contrast, KES-FB weight, extensibility, linearity of load, hysteresis of bending moment, shear stiffness, coefficient of friction, hysteresis of shear force at 5° shear angle, geometrical roughness, tensile resilience and mean deviation of MIU had p-value greater than 0.05, indicating no significant effect on the dependent variable.

Correlation and regression analysis between CLO fabric kit measurement results and drapability of silk fabrics

Results of correlation analysis between drape coefficient and CLO fabric kit properties are shown in Fig. 5. The drape ratio showed a significant positive correlation with bending stiffness (0.431**) (p-value < 0.01) and a significant negative correlation with weight

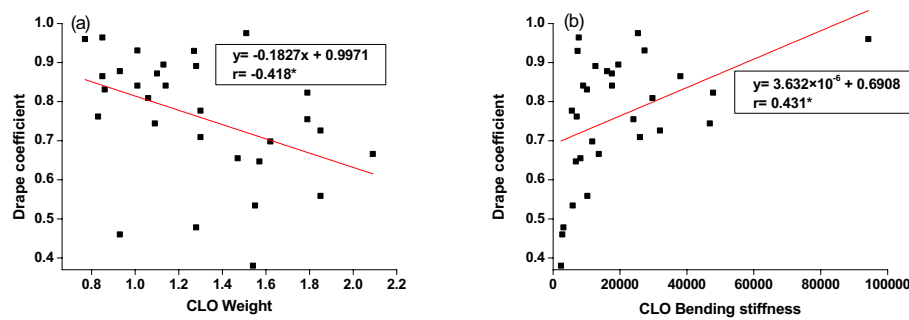


Fig. 5 Comparisons of drape coefficient and CLO fabric kit measurement values: **a** Weight, **b** Bending stiffness

(-0.418**). Similar to the correlation between KES-FB properties and drape coefficient, bending properties and weight showed similar correlations. However, tensile properties and shearing properties did not show significant correlations.

Table 5 shows results of multiple regression analysis between CLO fabric kit fabric properties and the drape coefficient, and the explanatory power (r^2) of the regression equation was 69.6%. Equation (4) is the regression equation of the regression model.

$$\begin{aligned} \text{Drape coefficient} = & 0.479 + 1.209 \times (\text{CLO Thickness}) - 0.254 \\ & \times (\text{CLO Weight}) + 1.476 \times 10^{-6} (\text{CLO Bending stiffness}) \end{aligned} \quad (4)$$

The drape coefficient is also expressed by bending property, thickness and weight with the CLO fabric kit results. As a result of checking the multicollinearity, CLO bending stiffness, thickness, and weight showed values of 0.1 or more in the case of tolerance and 10 or less in the case of variance inflation factor, indicating no problem of collinearity. In addition, the above three variables were found to have significant effects on the dependent variable (p -value < 0.05). On the other hand, CLO stretch stiffness and shear stiffness did not significantly affect the dependent variable (p -value > 0.05).

Comparison of drape coefficient between real and virtual silk fabrics

As a result of correlation analysis between drape coefficient of real fabric and the drape coefficient of virtual fabric, a high correlation of 0.824** was shown in Fig. 6.

Table 5 Regression analysis results of CLO fabric kit variables for drape coefficient of silk fabrics

Independent variables	Unstandardized coefficients		Standardized coefficients β	t	r^2
	B	Standard error			
(Constant)	0.479	0.122		1.649*	
CLO Thickness	1.209	0.411	0.360	2.940**	0.696
CLO Weight	-0.254	0.053	-0.580	-4.768***	
CLO Bending stiffness	1.476×10^{-6}	0.000	0.175	2.077*	

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

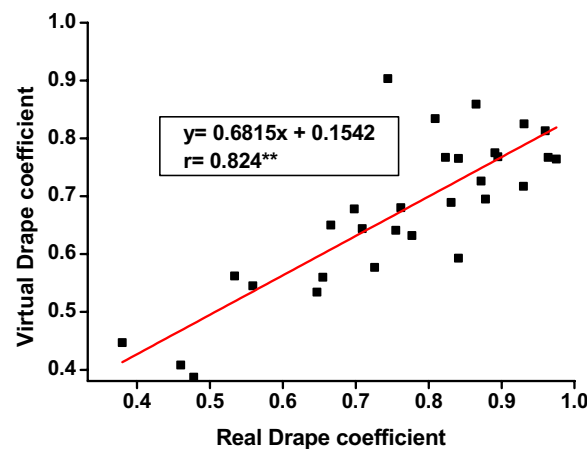


Fig. 6 Comparison between real and virtual drape coefficient of silk fabrics

Table 6 Independent samples t-test between real and virtual drape coefficient of silk fabrics

Drape coefficient	Mean	SD	Number of fabric	t
Real drape coefficient	0.7619	0.1564	30	5.330*
Virtual drape coefficient	0.6735	0.1294	30	

* $p < 0.05$

Although a high correlation was confirmed between the real and virtual drape coefficients, an independent t-test in SPSS was performed to determine whether there was a statistically significant difference between the real and virtual drape coefficients. At this time, the null hypothesis was set that there was no difference between the drape coefficient of the real fabric and the drape coefficient of the virtual fabric, and the alternative hypothesis was set that there was a difference between the two drape coefficients, and then the analysis was conducted. The analysis results showed that $t = 5.330$, $p = 0.03$ were statistically significant with a significance level of < 0.05 (Table 6). This means that there is a significant difference between the real and virtual drape coefficient. The real drape coefficient was found to be 0.7619 on average, and the virtual drape coefficient was 0.6735, showing a tendency for the virtual drape coefficient to be smaller for the same fabric. In the system that measures the tensile stiffness of the CLO fabric kit, the force gauge does not set the maximum force, but measures the stretched length and the force applied at that point by point. It is presumed that the resulting values, which did not sufficiently reflect the tensile strength of the stiff silk fabric used in this study, were reflected in the drape coefficient.

In order to group silk fabrics with similar drape coefficient (characteristics), K-means cluster analysis was performed on the actual drape coefficient, and as a result, they were classified into 4 grades. As shown in Table 7, grade A had the lowest drape coefficient. The drape coefficient increased as grades B, C, and D were classified into fabric groups. Figure 7 shows drape images and real and virtual drape profiles for three silk fabrics corresponding to the median value of each grade. Comparing the difference in drape coefficient by grade of silk fabric, the difference between the

real and virtual drape coefficient tended to increase as the grade increased. In other words, it was confirmed that the worse the drapability, that is, the stiffer the fabric, the larger the difference in drape coefficient between real and virtual, similar to Yoon's study (Yoon, 2021b). To analyze these results by grade, an independent sample t-test was conducted for real and virtual drape coefficient of silk fabrics belonging to each grade. As shown in Table 7, there was no statistically significant difference between real and virtual drape coefficients of fabrics belonging to grade A. There was a statistically significant difference between real and virtual drape coefficients of fabrics belonging to grades B, C, and D.

Table 7 Cluster analysis according to drape coefficient and independent sample t-test results between real & virtual drape coefficient of silk fabrics

Grade	Fabric No	Real drape coefficient	Virtual drape coefficient	Mean		Similarity of real and virtual coefficient	t
				Real drape coefficient	Virtual drape coefficient		
A	s19	0.380	0.477				
	s28	0.460	0.381	0.439	0.414	0.945	0.726
	s14	0.478	0.379				
B	s17	0.666	0.650				
	s12	0.647	0.534				
	s4	0.559	0.545	0.612	0.570	0.931	3.010**
	s10	0.534	0.562				
C	s4	0.655	0.560				
	s30	0.755	0.641				
	s11	0.841	0.593				
	s18	0.809	0.827				
	s24	0.726	0.668				
	s29	0.823	0.849	0.764	0.694	0.908	4.386***
	s22	0.762	0.775				
	s15	0.777	0.632				
	s23	0.698	0.678				
	s21	0.709	0.652				
	s8	0.744	0.891				
	s20	0.964	0.767				
	s9	0.930	0.717				
	s7	0.831	0.689				
	s16	0.895	0.821				
D	s6	0.872	0.726				
	s27	0.960	0.813	0.902	0.763	0.845	15.752***
	s25	0.841	0.765				
	s13	0.865	0.859				
	s3	0.891	0.775				
	s2	0.975	0.764				
	s26	0.931	0.825				
	s1	0.878	0.695				

** $p < 0.01$

*** $p < 0.001$

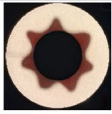
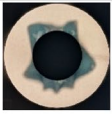
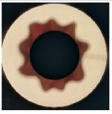
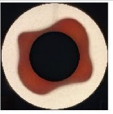
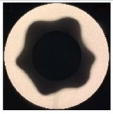
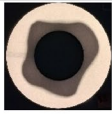
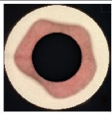
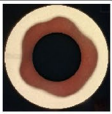
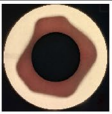
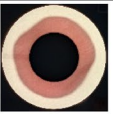
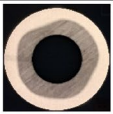
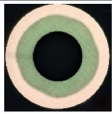
Grade	Grade A				Grade B	
Fabric No.	s22	s6	s17	s30	s23	s20
Real drape image						
Real drape profile						
Virtual drape profile						
Grade	Grade C			Grade D		
Fabric No.	s28	s2	s19	s13	s12	s8
Real drape image						
Real drape profile						
Virtual drape profile						

Fig. 7 Drape images and profiles of real and virtual silk fabrics

Looking at similarity of real and virtual drape coefficients according to grades divided by cluster analysis, the similarity between real and virtual drape coefficients of fabrics belonging to grade A was 94.5%, showing no statistical difference. However, it was found that there was a difference between real and virtual drape coefficients of silk fabrics belonging to grades B, C, and D. To analyze the cause of these results, physical properties of silk fabrics belonging to each grade were examined.

Table 8 shows weight, thickness, bending rigidity, and extensibility among KES-FB properties. These parameters showed high correlations with the drape coefficient. Silk fabrics belonging to grade A are plain weave fabrics woven with degummed silk yarns and patterned jacquard fabrics with a plain weave ground texture woven with degummed silk yarns. The bending rigidity had the smallest value of $0.089 \text{ gf}\cdot\text{cm}^2/\text{cm}$ and the extensibility had the highest value of 1.03%. Therefore, silk fabrics belonging to grade A were thin and flexible fabrics woven with degummed yarns having low fabric

density. It is known that if the bending rigidity value is small, it is easy to form a silhouette that is rich in elasticity, thus emphasizing curves of the body because it shows flexibility (Omeroglu et al., 2010). Lagé et al. (2020) have reported that the accuracy of the virtual try-on is significantly more useful than that of the actual garment when the bending rigidity of the fabric is less than 5.3 Nm (0.054 gf·cm²/cm) and the tensile strain in the warp direction is 1.80% or more similar to our results. The fabrics belonging to grades B and C were heavy in weight because they were woven with high warp and weft densities using degummed silk yarn. The jacquard fabrics and plain weave fabrics belonging to the middle group in thickness and bending property were groups with small values of extensibility. Fabrics belonging to grade D had the highest bending rigidity. They were mostly plain weave fabrics woven with thicker weft raw silk yarns.

Su'le (2012) has reported that the drape coefficient of fabrics with high bending stiffness was found to be high, which proved that the effect of bending stiffness of fabrics on drape properties is quite important. Therefore, when a virtual fabric was implemented with the CLO 3D virtual clothing program, the drape coefficient of the virtual fabric was realized almost similar to that of the real fabric for flexible and yielding fabrics. However, it was found that the drape coefficient difference between real and virtual fabrics was large for stiff and less stretchable fabrics. Lim and Istook (2011) have found that as flexural properties of fabrics increase, the stretch decreases and the drape becomes stiffer similar to results to the present study. When implementing virtual materials using the CLO 3D virtual clothing program, Lee and Kim (2011) have reported that the similarity between real fabrics and virtual fabrics with high warp and weft stiffness characteristics such as organza or thick and form-stable fabrics such as denim is generally low.

There is a lack of practical data for fabric properties in the 3D virtual clothing industry (Lee et al., 2011). To compensate for this, it is possible to express fabric properties in virtual space by applying physical properties measured in real fabrics. However, in addition to fabric properties provided by the virtual program, it is necessary to find appropriate values through various manual simulations (Chang & Lee, 2017). Based on the results of this study, virtual drape images were implemented for eight types of silk fabrics built in the fabric library in the CLO 7.1 program. As shown in Fig. 8, these eight types of virtual silk fabrics were taffeta, charmeuse, faille, crepe de-chine, double georgette, chiffon, duchess satin, and organza. Among them, taffeta, faille, and

Table 8 Physical properties of silk fabrics belonging to each grade

Grade	Mean/SD	Weight (mg/cm ²)	Thickness (mm)	Bending Rigidity (gf·cm ² /cm)	Extensibility (%)
A	Mean	6.653	0.233	0.089	1.03
	SD	1.598	0.014	0.015	0.139
B	Mean	9.159	0.290	0.241	0.636
	SD	1.596	0.075	0.084	0.081
C	Mean	7.515	0.279	0.271	0.594
	SD	1.983	0.070	0.097	0.079
D	Mean	5.577	0.297	0.331	0.673
	SD	1.101	0.105	0.086	0.150

organza showed large drape coefficient of 0.814, 0.822, and 0.872, respectively. Drape coefficient of the other five silk fabrics showed small values of 0.284 to 0.420. In other words, looking at the drape coefficient of silk fabric built into the CLO 3D program, the drapability of silk fabric in a low and high range can be realized. However, in the case of silk fabrics with a drape coefficient in the range of 0.5 to 0.7, there is a lack of actual silk fabric data base for use in implementing 3D virtual clothing. It seems necessary to secure a substantial data base that can accurately reflect the physical characteristics of silk fabric. In the virtual clothing simulation system currently in use, unlike the actual physical properties that have units, the bending and tensile property defined based on the particle system exist as dimensionless constant values without units, making it difficult to find a connection with the properties of the actual fabric. The three required fabric specimens through database creation.

Conclusions

The basic physical properties of fabrics were measured and conducted statistical analysis to investigate parameters affecting drapability for realizing 3D virtual clothing made of silk fabric. Additionally, the drapability of a virtual fabric implemented by the 3D CLO virtual clothing program and that of real fabrics were compared.

The drape coefficient of silk fabric was significantly correlated with extensibility, weight, tensile resilience, bending stiffness, and mean MIU deviation among KES-FB properties, and significantly correlated with bending stiffness and weight among CLO fabric kit measurements. The bending rigidity and weight per unit area were parameters influencing the drape coefficient.

Among the KES-FB properties, the tensile energy, bending stiffness, hysteresis width, thickness, and compression energy variables at a 0.5° shear angle predicted 78.6% of the drape coefficient. In addition, the drape coefficient could be predicted by 69.6% through

Silk Fabric	Taffeta	Charmeuse	Faille	Crepe de-chine
Drape coefficient	0.814	0.323	0.822	0.284
Virtual drape profile				
Silk Fabric	Double georgette	Chiffon	Duchess satin	Organza
Drape coefficient	0.313	0.420	0.247	0.872
Virtual drape profile				

Fig. 8 Drape profile and drape coefficient of virtual silk fabrics in 3D CLO fabric library

thickness, weight, and bending stiffness variables among the CLO fabric kit properties. It was found that bending is the variable that determines the drape coefficient of the silk fabric the most.

The physical property evaluation values of KES-FB and CLO fabric kits showed a high correlation in weight, thickness, and bending characteristics. However, no significant correlation was obtained between the two systems in terms of tensile properties. Therefore, it is necessary to adjust the physical properties of the program according to the tensile characteristics of the sample.

For stretchable and flexible silk fabrics, the real and virtual drape coefficients were in statistically significant agreement, confirming the possibility of clothing material implementation of the 3D digital clothing program. The CLO 7.1 program showed that the accuracy of virtual fabric was quite useful compared to real fabric if bending rigidity was lower than 0.089 gf·cm²/cm and extensibility was higher than 1.03%. However, it also confirmed a difference in similarity between real and virtual silk fabrics, which had a large drape coefficient of 0.6 or more with stiffness. In the case of stiff fabrics with low elasticity and high bending properties, more research is needed on the program modeling method considering the mechanical and physical properties of real fabric and its structure.

Author contributions

JL proposed the research idea and JHK carried out the research. JHK were mainly responsible for data analysis and the experiment along with writing the manuscript, and JL was involved in the edition of the manuscript as well as data analysis and the experiments. All of the authors read and approved the final version of the manuscript.

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Availability of data and materials

The datasets used and/or analysed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate

Not applicable.

Competing interests

The author declares no competing interests.

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